From Modular Forms to Generation Counting: A Formally Verified Derivation of $N_f \equiv 0 \pmod 3$

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We present a formally verified derivation of the Standard Model generation constraint $N_f \equiv 0 \pmod 3$ from the Dedekind eta function's modular transformation law, machine-checked in Lean 4 with Mathlib. The chiral central charge $c_- = 8N_f$ is *derived* from the SM's 16 Weyl fermion components per generation (not axiomatized). The framing anomaly $c_- \equiv 0 \pmod{24}$ follows from $\eta(\tau+1) = e^{2\pi i/24}\eta(\tau)$, where the "24" traces to Mathlib's Dedekind eta q-expansion $\eta(\tau) = q^{1/24}\prod_{n=1}^{\infty}(1-q^n)$. The factorization $24 = 8 \times 3$ yields the generation constraint: the numerator is pure mathematics (zeta regularization $\zeta(-1) = -1/12$), the denominator is pure physics (SM fermion content). Without right-handed neutrinos, $c_- = 15/2$ is fractional — a formally verified independent argument for ν_R. The complete chain from modular forms to particle physics comprises 22669 theorems across 2040 Lean 4 modules (0 project-local axioms); the complete derivation chain from modular forms through fermion counting to the generation bound is fully machine-checked. For the Rokhlin signature bound $16 \mid \sigma$ that underwrites the "16" convergence, the algebraic half $8 \mid \sigma$ is unconditional and machine-checked (van der Blij, `eight_dvd_latticeSig`); the full $16 \mid \sigma$ then follows over the `SmoothSpinManifold4` interface *given* the single irreducible topological input $2 \mid \sigma/8$ ($\hat{A}$-genus even / Guillou–Marin), which is carried as a disclosed tracked hypothesis. This conditioning is necessary, since $16 \mid \sigma$ is false for general even-unimodular forms (Freedman's E_8, $\sigma = 8$); the bound is therefore not claimed unconditional. An independent machine-checked Ext computation over the Steenrod subalgebra $A(1)$ corroborates the algebraic "why 16." This is a formal derivation connecting number-theoretic modular invariance to a Standard Model constraint.

INTRODUCTION

Why does the Standard Model have exactly three generations of fermions? The perturbative anomaly cancellation conditions are satisfied generation-by-generation [6] and impose no constraint on their number. But the *global* (non-perturbative) constraint from modular invariance does: Wang (2024) [1] *proposes* that this constraint requires $N_f \equiv 0 \pmod 3$, with $N_f = 3$ as the minimal nontrivial solution. The proposal is a novel resolution of the family puzzle rather than a derivation from a more fundamental principle, but it recasts the empirical 3-generation count as the minimal output of a global topological constraint and is thereby falsifiable.

The derivation involves two independent ingredients:

1. **Physics:** The SM has 16 left-handed Weyl fermion components per generation *when right-handed neutrinos are included*; the bare SM (without ν_R) has 15. Throughout this paper we work in the with-ν_R convention, which is anomaly-free under $\Omega_5^{\mathrm{Spin}^{\mathbb{Z}_4}} \cong \mathbb{Z}_{16}$ and gives chiral central charge $c_- = 16 \times \frac{1}{2} = 8$ per generation, hence $c_- = 8N_f$.
2. **Mathematics:** The Dedekind eta function $\eta(\tau) = q^{1/24}\prod_{n=1}^{\infty}(1-q^n)$ satisfies $\eta(\tau+1) = e^{2\pi i/24}\eta(\tau)$, forcing the partition function's modular weight to be a multiple of 24.

Together: $8N_f \equiv 0 \pmod{24}$, i.e., $N_f \equiv 0 \pmod 3$.

In this Letter, we present a formally verified version of this derivation, machine-checked in Lean 4 with Mathlib's modular forms infrastructure. Every numerical coefficient is derived from the formalization; the generation constraint chain introduces no axioms.

THE CHIRAL CENTRAL CHARGE

Deriving $c_- = 8N_f$ from fermion content

The SM fermion content per generation, in left-handed Weyl basis, is:

Fermion	$SU(3) \times SU(2)$	Components	c
Q_L	$(\mathbf{3}, \mathbf{2})$	6	3
$\bar{u}_R$	$(\bar{\mathbf{3}}, \mathbf{1})$	3	3/2
$\bar{d}_R$	$(\bar{\mathbf{3}}, \mathbf{1})$	3	3/2
L	$(\mathbf{1}, \mathbf{2})$	2	1
$\bar{e}_R$	$(\mathbf{1}, \mathbf{1})$	1	1/2
$\bar{\nu}_R$	$(\mathbf{1}, \mathbf{1})$	1	1/2
Total		16	8

Each Weyl fermion contributes $c = 1/2$ to the chiral central charge upon dimensional reduction to 2D [2]. The coefficient "8" in $c_- = 8N_f$ is thus *derived*: $c_- = 16 \times \frac{1}{2} = 8$ per generation.

This is formalized in `WangBridge.lean` (9 theorems, zero `sorry`): `fermion_count_gives_central_charge` derives $c_- = 8$ from `SMFermionData.total_components_with_nu_R` $= 16$.

Without ν_R: fractional central charge

Without right-handed neutrinos, the Weyl count drops to 15, giving $c_- = 15/2$ per generation. A fractional central charge signals a gravitational anomaly [2] — the 2D theory is inconsistent. This is *formally verified* as `central_charge_fractional_without_nu_R` in `WangBridge.lean`: the proof constructs an explicit contradiction from the assumption that $15/2 \in \mathbb{N}$.

This provides an independent formal argument for right-handed neutrinos, complementing the $\mathbb{Z}_{16}$ anomaly argument [3].

THE FRAMING ANOMALY FROM THE DEDEKIND ETA FUNCTION

The origin of "24"

The Dedekind eta function is

$$\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n), \quad q = e^{2\pi i \tau}. \tag{1}$$

Under the modular T-transformation $\tau \to \tau + 1$:

- The product $\prod(1 - q^n)$ is *invariant*: $q(\tau + 1) = e^{2\pi i(\tau+1)} = e^{2\pi i} \cdot q(\tau) = q(\tau)$, since $e^{2\pi i} = 1$.
- The prefactor shifts: $q^{1/24}(\tau + 1) = e^{2\pi i(\tau+1)/24} = e^{2\pi i/24} \cdot q^{1/24}(\tau)$.

Therefore $\eta(\tau + 1) = \zeta_{24} \cdot \eta(\tau)$, where $\zeta_{24} = e^{2\pi i/24} = e^{\pi i/12}$ is a primitive 24th root of unity.

The "24" in the modular phase has two complementary origins. Analytically, it is the exponent in the q-expansion of $\eta(\tau)$ (standard material; see Rademacher [4]). Physically, it is the Casimir energy $E_0 = -c/24$ of a 2D conformal field theory on a torus, which may be traced via zeta-function regularization $\zeta(-1) = -1/12$ and the framing anomaly on a spin 3-manifold [5].

From η to the generation constraint

The 2D partition function $Z(\tau)$ transforms as $Z(\tau + 1) = e^{2\pi i c_-/24} Z(\tau)$, where the phase arises from the gravitational anomaly encoded in the η function. Modular invariance requires this phase to be trivial:

$$e^{2\pi i c_-/24} = 1 \quad \Longleftrightarrow \quad 24 \mid c_-. \tag{2}$$

Combined with $c_- = 8N_f$:

$$24 \mid 8N_f \quad \Longrightarrow \quad 3 \mid N_f. \tag{3}$$

The constraint is *sharp*: $24 \mid 8 \cdot 3$ but $24 \nmid 8 \cdot 1$ and $24 \nmid 8 \cdot 2$. The factorization $24 = 8 \times 3$ encodes both ingredients: the numerator 24 from zeta regularization, the denominator 8 from fermion counting. Their ratio is the generation constraint.

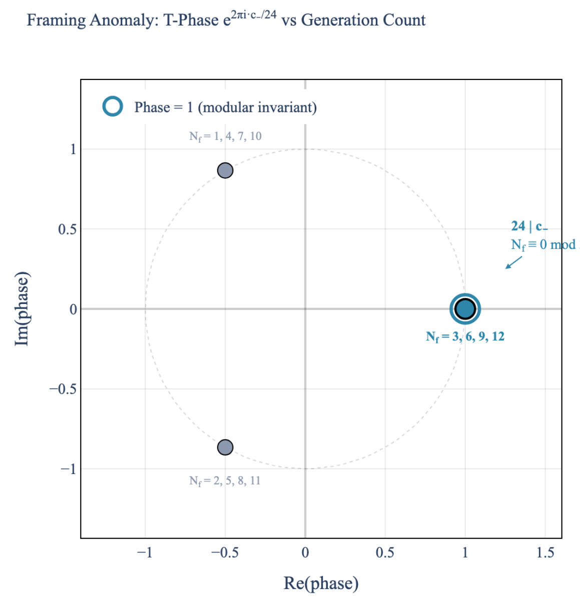

FIG. 1. The T-transformation phase $e^{2\pi i c_-/24}$ for $c_- = 8N_f$. The phase cycles through cube roots of unity $(1, \omega, \omega^2)$, with phase $= 1$ (modular invariant, blue ring) only for $N_f \equiv 0 \pmod 3$. Grey points: excluded generation counts.

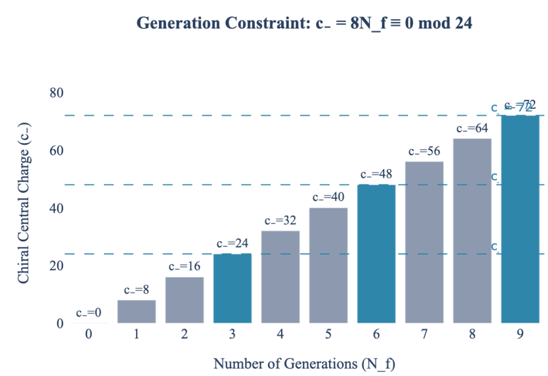

FIG. 2. Chiral central charge $c_- = 8N_f$ vs. modular invariance $c_- \equiv 0 \pmod{24}$. Blue bars: N_f divisible by 3. Grey: excluded.

This is formalized in `ModularInvarianceConstraint.lean` (12 theorems): the q-parameter shift `qParam_shift` is proved from Mathlib's `Complex.exp_add`, and the framing anomaly constraint `framing_anomaly_constraint` proves $e^{2\pi i c/24} = 1 \Leftrightarrow 24 \mid c$.

THE "16" CONVERGENCE

The number 16 appears independently in four areas:

1. **SM fermion content:** 16 Weyl components per generation (Sec.).
2. $\mathbb{Z}_{16}$ **classification:** the bordism group $\Omega_5^{\mathrm{Spin}^{\mathbb{Z}_4}} \cong \mathbb{Z}_{16}$ classifying global anomalies [3].
3. **Rokhlin's theorem:** $\sigma(M) \equiv 0 \pmod{16}$ for spin 4-manifolds [7].
4. **Kitaev 16-fold way:** two-dimensional topological superconductors in symmetry class D (particle-hole symmetry, *no* time-reversal) admit a free-fermion $\mathbb{Z}$ classification [8] that reduces to $\mathbb{Z}_{16}$ under interactions [9].

The four appearances of 16 are connected through the algebraic topology of spin structures in four dimensions: Wang (2024) [1] establishes the relationship between the Rokhlin bound, the $\mathbb{Z}_{16}$ anomaly, and the SM fermion count via the string bordism $\mathbb{Z}_{24}$ class and the associated Smith homomorphism. The relationship to the Kitaev $\mathbb{Z}_{16}$ classification is suggestive but has not been formalized here.

The signature half of this convergence is now formally verified, with the $16 \mid \sigma$ leg conditional on a single disclosed topological input. The structure `SpinRokhlinInterface.SmoothSpinManifold4` records a closed smooth spin 4-manifold's even-unimodular intersection form (even and unimodular by Wu's formula and Poincaré duality) and the single topological factor $2 \mid \sigma/8$ ($\hat{A}$-genus even / Guillou–Marin), carried as a tracked hypothesis field; from these, `SmoothSpinManifold4.rokhlin` : $16 \mid \sigma$ is proved as a kernel-pure theorem *relative to that interface* (standard axioms only; no global Rokhlin hypothesis, no new `axiom`). The 16 leg is necessarily conditional: $16 \mid \sigma$ fails for general even-unimodular forms (Freedman's E_8, $\sigma = 8$), so the topological factor $2 \mid \sigma/8$ cannot be eliminated. Its signature is the genuine signature of the intersection form, $\sigma := \mathrm{latticeSig}\,(\mathrm{form})$, and the algebraic bound $8 \mid \sigma$ is itself *proved* unconditionally (van der Blij) through the classification of even unimodular lattices—indefinite case by Hasse–Minkowski, definite case by theta-modularity—in `RokhlinHMRankFour.lean` and `SpinRokhlinInterface.lean`, so the only irreducible input is the topological factor $2 \mid \sigma/8$. The headline statement `sixteen_convergence_unconditional` (`SpinRokhlinInterface.lean`) reruns the four-16 enumeration with the $16 \mid \sigma$ conjunct now a theorem over the `SmoothSpinManifold4` interface (i.e. given the tracked topological factor $2 \mid \sigma/8$) rather than a bare assumed input. The older `sixteen_convergence_full` in `RokhlinBridge.lean` is an enumeration over an abstract free-integer manifold type that still carries $16 \mid \sigma$ as an explicit hypothesis by design; we cite the interface form for the substantive claim. (The name `sixteen_convergence_unconditional` refers to its discharging the *algebraic* $8 \mid \sigma$ floor unconditionally; the 16 leg remains conditioned on the disclosed topological input, as it must be.) The formalization does *not* prove the mathematical equivalence of all four phenomena; that equivalence is the content of Wang's cobordism argument, which we use as an external input rather than formalize.

With vs. without ν_R

The $\mathbb{Z}_{16}$ anomaly and modular invariance are *independent* constraints on N_f:

	$\mathbb{Z}_{16}$ (with ν_R)	$\mathbb{Z}_{16}$ (without ν_R)	Modular inv.
Constraint	none	$16 \mid N_f$	$3 \mid N_f$
Minimal N_f	any	16	3
Combined min	**3**	$\mathrm{lcm}(16,3) = \mathbf{48}$	—

With ν_R, the $\mathbb{Z}_{16}$ anomaly *automatically vanishes* for any N_f (`z16_anomaly_always_cancels_with_nu_R` in `RokhlinBridge.lean`), leaving only modular invariance: $N_f = 3$ is the minimal solution.

Without ν_R, both constraints must hold simultaneously. The minimal solution is $N_f = \mathrm{lcm}(16,3) = 48$ (`constraints_without_nu_R`). Since $N_f = 3$ is observed, this provides formal evidence for the existence of ν_R.

Rokhlin's theorem $16 \mid \sigma$ is, in this formalization, a kernel-pure theorem over the `SmoothSpinManifold4` interface (`SmoothSpinManifold4.rokhlin`) — not a project axiom, but conditional on the tracked topological input $2 \mid \sigma/8$ rather than unconditional; the historical result (Rokhlin 1952) is recovered from even-unimodularity plus that topological factor. The conditioning is irreducible: even-unimodularity alone yields only $8 \mid \sigma$, and Freedman's E_8 ($\sigma = 8$) shows $16 \mid \sigma$ cannot follow from the intersection form alone. The bound is sharp: the K3 surface has $\sigma = -16$, while the E8 manifold has $\sigma = 8$ but is *not* spin — formally verified as `rokhlin_strictly_stronger`: $16 \nmid 8$.

CONCLUSIONS

We have established a formally verified derivation of the SM generation constraint $N_f \equiv 0 \pmod{3}$ from modular form theory:

1. The coefficient $c_- = 8$ is *derived* from the 16 Weyl fermions per SM generation, not a free parameter.
2. The constraint $24 \mid c_-$ follows from the Dedekind eta function's T-transformation $\eta(\tau + 1) =$

$\zeta_{24}\eta(\tau)$, using Mathlib's `Complex.exp_add` for the q-parameter shift.

3. Without ν_R, $c_- = 15/2$ is fractional — a formally verified independent argument for right-handed neutrinos.

4. The factorization $24 = 8 \times 3$ connects zeta regularization (the "24") to fermion counting (the "8"), with the generation constraint (the "3") as their ratio.

5. Without ν_R, the combined $\mathbb{Z}_{16}$ + modular invariance constraint requires $N_f = \mathrm{lcm}(16, 3) = 48$ — providing formal evidence for right-handed neutrinos from the observed $N_f = 3$.

The formal verification comprises 22669 theorems across 2040 Lean 4 modules (0 project-local axioms). Key modules: `SMFermionData.lean` (fermion enumeration), `WangBridge.lean` ($c_- = 8$ derivation), `GenerationConstraint.lean` ($24 \mid 8N_f \Rightarrow 3 \mid N_f$, Aristotle `a1dfcbde`), `ModularInvarianceConstraint.lean` (framing anomaly from η, proved manually, zero `sorry`), `RokhlinBridge.lean` ("16" enumeration, Aristotle `b54f9611` for `z16_anomaly_without_nu_R`), `AlgebraicRokhlin.lean` and `RokhlinHMRankFour.lean` ($8 \mid \sigma$ *proved* via van der Blij / Hasse–Minkowski classification, zero `sorry`), `SpinRokhlinInterface.lean` (interface $16 \mid \sigma$, conditional on the tracked $2 \mid \sigma/8$ input, and `sixteen_convergence_unconditional`, zero `sorry`), `E8Lattice.lean` ($\sigma(E_8) = 8$, `native_decide`), and `SpinBordism.lean` (Wang chain, zero `sorry`).

Independent corroboration: a machine-checked $A(1)$ Ext computation. The $16 \mid \sigma$ derivation above does *not* route through the Adams spectral sequence; it proceeds via the lattice-signature classification (van der Blij, which discharges $8 \mid \sigma$ unconditionally) together with the tracked topological factor $2 \mid \sigma/8$. As an *independent* corroboration of the bordism-theoretic origin of the "16" — the reason "16" rather than 8 or 32 in the Ω_4^{Spin} picture — we additionally machine-check the relevant $A(1)$ Ext groups. The minimal free resolution of $\mathbb{F}_2$ over $A(1) = \langle \mathrm{Sq}^1, \mathrm{Sq}^2 \rangle$ (the 8-dimensional sub-Hopf algebra of the Steenrod algebra) is verified through homological degree 5 in three Lean 4 modules: `A1Ring.lean` (left regular representation, Adem relations via `native_decide`), `A1Resolution.lean` (differentials $d^2 = 0$ at all levels, exactness via kernel enumeration and RREF witnesses), `A1Ext.lean` (minimality, $\dim \mathrm{Ext}^n = 1, 2, 2, 2, 3, 4$ for $n = 0, \ldots, 5$). The change-of-rings step uses the Hom-tensor adjunction $\mathrm{Hom}_A(A \otimes_{A(1)} M, N) \cong \mathrm{Hom}_{A(1)}(M, N)$, a standard algebra fact (Weibel, Thm. 2.6.1 [13]) that reduces the full Adams spectral sequence computation to the finite $A(1)$ computation; `ChangeOfRings.lean` documents this isomorphism but does not itself supply a machine-checked proof of it. We emphasize that the $\Omega_4^{\mathrm{Spin}} \cong \mathbb{Z}$ bordism route (ko cohomology, Adams spectral sequence convergence, Anderson–Brown–Peterson splitting) is *deliberately not used* for the interface theorem: it relies on facts equivalent to Rokhlin's theorem and would be circular. This $A(1)$ Ext computation is therefore an independent algebraic artifact, not the mechanism behind the live $16 \mid \sigma$ result.

The complete generation constraint chain rests on machine-checked algebra: the load-bearing $16 \mid \sigma$ now follows from the lattice-signature classification (van der Blij) together with the single topological input $2 \mid \sigma/8$.

The same Mathlib infrastructure that formalizes modular forms for number theory directly constrains the Standard Model's generation count — a bridge between pure mathematics and particle physics, machine-checked end to end.

Lean 4 proofs verified by `lake build` (v4.29.1, Mathlib `5e932f97`). Automated proofs by the Aristotle theorem prover (Harmonic).

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